

ING Credit Risk: formula sheet

Prep for Product Owner & Senior Risk Data Journey Expert · Credit Infrastructure

Standalone crib for Anjish Bhondwe. Basel IRB / IFRS 9 textbook forms. Not ING policy. Also included in the full work simulation PDF.

MUST KNOW: $EL = PD \times EAD \times LGD$

Expected Loss = Probability of Default $\times$ Exposure at Default $\times$ Loss Given Default

Multiply. Do not add. PD not "PL".

16. Formula sheet (all key equations)

One-page crib. Multiply where shown. PD = Probability of Default (not "PL"). Detail and worked numbers follow on the next pages.

Loss building blocks

$$EL = PD \times EAD \times LGD$$

$$EL\% = PD \times LGD$$

$$EAD = \text{Drawn} + CCF \times \text{Undrawn} \quad \text{with} \quad \text{Undrawn} = \text{Limit} - \text{Drawn}$$

$$LGD = 1 - RR \quad \cdot \quad RR = NPV(\text{Recoveries} - \text{Costs}) / EAD$$

$$CCF = (\text{Bal_default} - \text{Bal_ref}) / \text{Undrawn_ref} \quad (0 \text{ to } 1, \text{ capped})$$

$$PD = 1 / (1 + e^{-z}) \quad (\text{logit score example; } z = \beta_0 + \beta \cdot x)$$

$$PD_{\text{port}} = \sum (PD_i \times EAD_i) / \sum EAD_i$$

Capital / RWA (IRB)

$$R = 0.12 \times (1 - e^{-50 \times PD}) / (1 - e^{-50}) + 0.24 \times [1 - (1 - e^{-50 \times PD}) / (1 - e^{-50})]$$

$$b = (0.11852 - 0.05478 \times \ln(PD))^2$$

$$K = [LGD \times N(G(PD) / \sqrt{(1-R)}) + \sqrt{(R/(1-R))} \times G(0.999)] - PD \times LGD \times (1 + (M-2.5)b) / (1-1.5b)$$

$$RWA = 12.5 \times K \times EAD \quad \cdot \quad \text{Capital} \approx K \times EAD \quad \cdot \quad \text{CET1 ratio} = \text{CET1} / RWA$$

$$\text{Standardised: } RWA = \text{RiskWeight} \times \text{Exposure}$$

$$\text{Retail mortgage (common): } R = 0.15 \text{ (fixed)}$$

IFRS 9 ECL / LLP

$$\text{Stage 1: } ECL_{12m} \approx PD_{12m} \times LGD \times EAD \times DF$$

$$\text{Stage 2/3: } ECL_{LT} = \sum_t [PD_{\text{marg},t} \times LGD_t \times EAD_t / (1+r)^t]$$

$$PD_{\text{marg},t} = \text{Survival}_{\{t-1\}} \times PD_{\text{cond},t} \quad \cdot \quad \text{Survival}_t = \prod_{\{s \leq t\}} (1 - PD_{\text{cond},s})$$

$$\text{Scenarios: } ECL = \sum_s w_s \times ECL_s \quad (\sum w_s = 1)$$

$$LLP \text{ charge} \approx ECL_{\text{end}} - ECL_{\text{start}} + \text{write-offs} - \text{recoveries}$$

Rule engines and lending ratios

$$\text{DoD} = 1 \text{ if } (DPD \geq 90) \text{ OR UTP OR specific forbearance (else 0)}$$

$$DPD = \text{ReportDate} - \text{OldestUnpaidDueDate}$$

$$EWS_score = \sum w_i \times \text{signal}_i \quad ; \text{ flag if score } \geq \text{threshold}$$

$$LTV = \text{Loan (or EAD)} / \text{Collateral_value}$$

$$DSTI = \text{Debt_service} / \text{Income} \quad (\text{same period})$$

$$DTAI = \text{Total_debt} / \text{Total_assets} \quad \text{OR} \quad \text{Debt_service} / \text{Available_income} \quad (\text{confirm ING})$$

$$\text{Utilisation} = \text{Drawn} / \text{Limit}$$

$$\text{Collateral coverage} = \text{Collateral_value} \times (1 - \text{Haircut}) / EAD$$

$$\text{NPL ratio} = \text{NPL_gross} / \text{Total_loans_gross}$$

$$\text{Provision coverage} = \text{LLP_stock} / \text{NPL_gross}$$

Numeric chain (same example everywhere)

$$\text{Limit } 100k, \text{ Drawn } 80k, \text{ CCF } 50\% \rightarrow EAD = 90k$$

$$PD \text{ } 2\%, LGD \text{ } 40\% \rightarrow EL = 0.02 \times 90k \times 0.40 = 720$$

$$LTV \text{ on } 90k / 120k \text{ property} = 75\% \cdot DSTI \text{ } 1,200 / 4,000 = 30\%$$

$$IRB (M=2.5): R \approx 0.164 \cdot K \approx 8.17\% \rightarrow RWA \approx 91.9k \cdot \text{capital} \approx 7.35k$$

16a. Formula detail: core parameters

Same equations as the formula sheet, with definitions. Not bank policy

A. Core parameters (what your engines produce)

Term	Definition	Full formula
PD	Probability of default over horizon T (IRB often 1 year; IFRS 9 Stage 2/3 uses lifetime).	$0 \leq PD \leq 1$ Rating model (logit example): $PD = 1 / (1 + e^{-z})$ where $z = \beta_0 + \beta_1 x_1 + \dots$ Exposure-weighted portfolio PD: $PD_{port} = \sum (PD_i \times EAD_i) / \sum EAD_i$
CCF	Share of undrawn limit expected to be drawn at default.	$0 \leq CCF \leq 1$ Historical: $CCF = (\text{Balance_at_default} - \text{Balance_at_ref}) / \text{Undrawn_at_ref}$ (capped)
EAD	Exposure if default occurs.	$EAD = \text{Drawn} + CCF \times \text{Undrawn}$ where Undrawn = Limit - Drawn (netting, collateral substitution, product caps may adjust)
LGD	Loss fraction of EAD after recoveries, costs, discounting.	$LGD = 1 - RR$ $RR = NPV(\text{Recoveries} - \text{Costs}) / EAD$ Workout path example: $LGD = [EAD - \sum_t (\text{Cash}_t - \text{Cost}_t) / (1+r)^t] / EAD$ Secured intuition: higher effective collateral → lower LGD
EL Expected Loss	Average credit loss over the horizon. Building block for LLP/ECL thinking. Capital covers unexpected loss above EL.	$EL = PD \times EAD \times LGD$ Example: $0.02 \times 90,000 \times 0.40 = 720$ $EL\% = PD \times LGD = 0.02 \times 0.40 = 0.80\%$ of EAD Wrong form: $EL \neq PD + LGD + EAD$
UL	Unexpected loss. IRB K approximates a high-percentile UL charge.	Concept: $UL \approx \text{Loss_at_99.9\%} - EL$ IRB uses formula for K below (not a simple σ formula in retail decks)

B. Worked mini-example (memorise the arithmetic)

Drawn = 80,000 · Limit = 100,000 · Undrawn = 20,000 · CCF = 0.50

$EAD = 80,000 + 0.50 \times 20,000 = 90,000$

PD = 2% = 0.02 · LGD = 40% = 0.40

$EL = 0.02 \times 90,000 \times 0.40 = 720$

$EL\% = 0.02 \times 0.40 = 0.80\%$ of EAD

C. Capital, RWA: full Basel IRB (ASRF) formulas

Corporate / bank / sovereign style IRB. $N(\cdot)$ = standard normal CDF. $G(\cdot)$ = inverse standard normal. M = effective maturity in years. Retail asset classes use different correlation formulas (see note).

1) Asset correlation R (corporate):

$$R = 0.12 \times (1 - e^{-50 \times PD}) / (1 - e^{-50}) + 0.24 \times [1 - (1 - e^{-50 \times PD}) / (1 - e^{-50})]$$

2) Maturity adjustment b :

$$b = (0.11852 - 0.05478 \times \ln(PD))^2$$

3) Capital requirement K per unit EAD:

$$K = [LGD \times N((1-R)^{-0.5} \times G(PD) + (R/(1-R))^{0.5} \times G(0.999)) - PD \times LGD] \times (1 + (M - 2.5) \times b) / (1 - 1.5 \times b)$$

4) RWA and capital:

$$RWA = 12.5 \times K \times EAD \quad (\text{because } 12.5 = 1 / 8\%)$$

$$\text{Required capital} \approx K \times EAD \quad (\text{at } 8\% \text{ of RWA})$$

$$\text{CET1 ratio} = \text{CET1 capital} / RWA$$

Retail mortgage correlation (common IRB variant): $R = 0.15$ (fixed in many Basel retail-mortgage specs; other retail classes use PD-dependent R). Same K structure without always applying the corporate maturity term the same way. Check asset class.

Standardised approach (contrast): $RWA = \text{RiskWeight} \times \text{Exposure}$. RiskWeight from tables (e.g. by rating or LTV bucket), not from PD/LGD models.

D. Worked IRB example ($PD = 2\%$, $LGD = 40\%$, $M = 2.5$, $EAD = 90,000$)

From formula (1): $R \approx 0.1641$

From formula (2): $b \approx 0.1108$ (maturity factor = 1 when $M = 2.5$)

From formula (3): $K \approx 0.0817$ (about 8.17% of EAD)

$$RWA = 12.5 \times 0.0817 \times 90,000 \approx 91,883$$

$$\text{Required capital} \approx K \times EAD \approx 7,351$$

Compare EL = 720 (much smaller than UL-style capital charge at 99.9%)

E. IFRS 9 ECL / LLP formulas

Item	Formula
Stage 1 (12-month ECL)	$ECL_{12m} \approx PD_{12m} \times LGD \times EAD \times DF$ DF = discount factor for expected timing of loss
Stage 2 / 3 (lifetime ECL)	$ECL_{LT} = \sum_{t=1..T} [PD_{marg,t} \times LGD_t \times EAD_t / (1+r)^t]$ $PD_{marg,t} = survival_{t-1} \times PD_{cond,t}$ $Survival_t = \prod_{s \leq t} (1 - PD_{cond,s})$
Scenario-weighted	$ECL = \sum_s w_s \times ECL_s$ with $\sum w_s = 1$ (e.g. base / up / down)
LLP / allowance	Stock of credit loss allowance $\approx$ aggregated ECL (plus management overlays if any) $LLP \text{ charge (P\&L)} \approx ECL_{end} - ECL_{start} + \text{write-offs} - \text{recoveries}$ (simplified)

DoD flags and SICR (significant increase in credit risk) drive staging. Staging $\neq$ DoD, but DoD usually forces Stage 3.

F. Rule engines and lending indicators (explicit calcs)

Term	Definition	Calculation
DoD	Definition of Default: rule engine marking default.	DoD = 1 if any trigger true, else 0 Typical triggers (policy-specific): $DPD \geq 90$ OR $UTP = 1$ OR specific forbearance Obligor default may contagion to all facilities
DPD	Days past due.	DPD = ReportDate - OldestUnpaidDueDate (product rules / grace may apply)
UTP	Unlikely to pay (judgemental default).	$UTP = 1$ if policy indicators fire (distress, bankruptcy filing, etc.). No single closed formula.
EWS	Early warning before DoD.	Score example: $EWS_score = \sum w_i \times signal_i$ Flag if $EWS_score \geq$ threshold (set with Credit Risk) Signals: arrears trend, utilisation $\uparrow$, covenant breach, income drop
LTV	Loan to value.	LTV = Outstanding_loan (or EAD) / Collateral_value Example: 270,000 / 300,000 = 0.90 (90%)
DSTI	Debt service to income.	DSTI = Monthly_debt_service / Monthly_income Example: 1,400 / 4,000 = 0.35 (35%)
DTAI	Confirm ING label. Common: debt-to-assets or debt-to-available-income.	Debt-to-assets: DTAI = Total_debt / Total_assets Debt-to-available-income: DTAI = Debt_service / Available_income $Available_income = Income - taxes - essential\ living$ (policy definition)
Utilisation	How much of limit is drawn.	Utilisation = Drawn / Limit
Collateral coverage	Cover after haircut.	Coverage = (Collateral_value $\times$ (1 - Haircut)) / EAD
NPL ratio	Non-performing stock vs book.	NPL ratio = NPL_gross / Total_loans_gross
Coverage ratio (provisions)	Allowance vs NPL.	Prov. coverage = LLP_stock / NPL_gross

G. End-to-end numeric chain (interview whiteboard)

Limit 100k, Drawn 80k, CCF 50% $\rightarrow$ **EAD = 90k**
 PD 2%, LGD 40% $\rightarrow$ **EL = 0.02 $\times$ 90k $\times$ 0.40 = 720**
 Collateral value 120k, haircut 20% $\rightarrow$ Cover = 120k $\times$ 0.8 / 90k = **1.067**
 LTV on 90k vs 120k property = **75%**
 Income 4,000/mo, debt service 1,200 $\rightarrow$ **DSTI = 30%**
 If K = 8.17% (from worked IRB example) $\rightarrow$ **RWA $\approx$ 12.5 $\times$ 0.0817 $\times$ 90k $\approx$ 91.9k** · capital $\approx$ **7.35k**
 Stage 1 ECL (ignore DF): $\approx$ **PD12m $\times$ LGD $\times$ EAD = 720** (same building blocks; timing/scenarios adjust)

H. Term index (quick definitions)

IRB = Internal Ratings-Based · IFRS 9 = ECL accounting · CRR/CRD = EU capital law · AnaCredit = ECB loan-level reporting · Forbearance = concession to distressed client · Cure = return to performing · Record Bank residual = legacy codes in BE landscape · SICR = significant increase in credit risk (Stage 2 trigger).

End of formula sheet. Pair with: anjish-bhondwe-ing-credit-infrastructure-work-simulation.pdf and CV anjish-bhondwe-cv-ing-po-risk-data-journey-expert.pdf